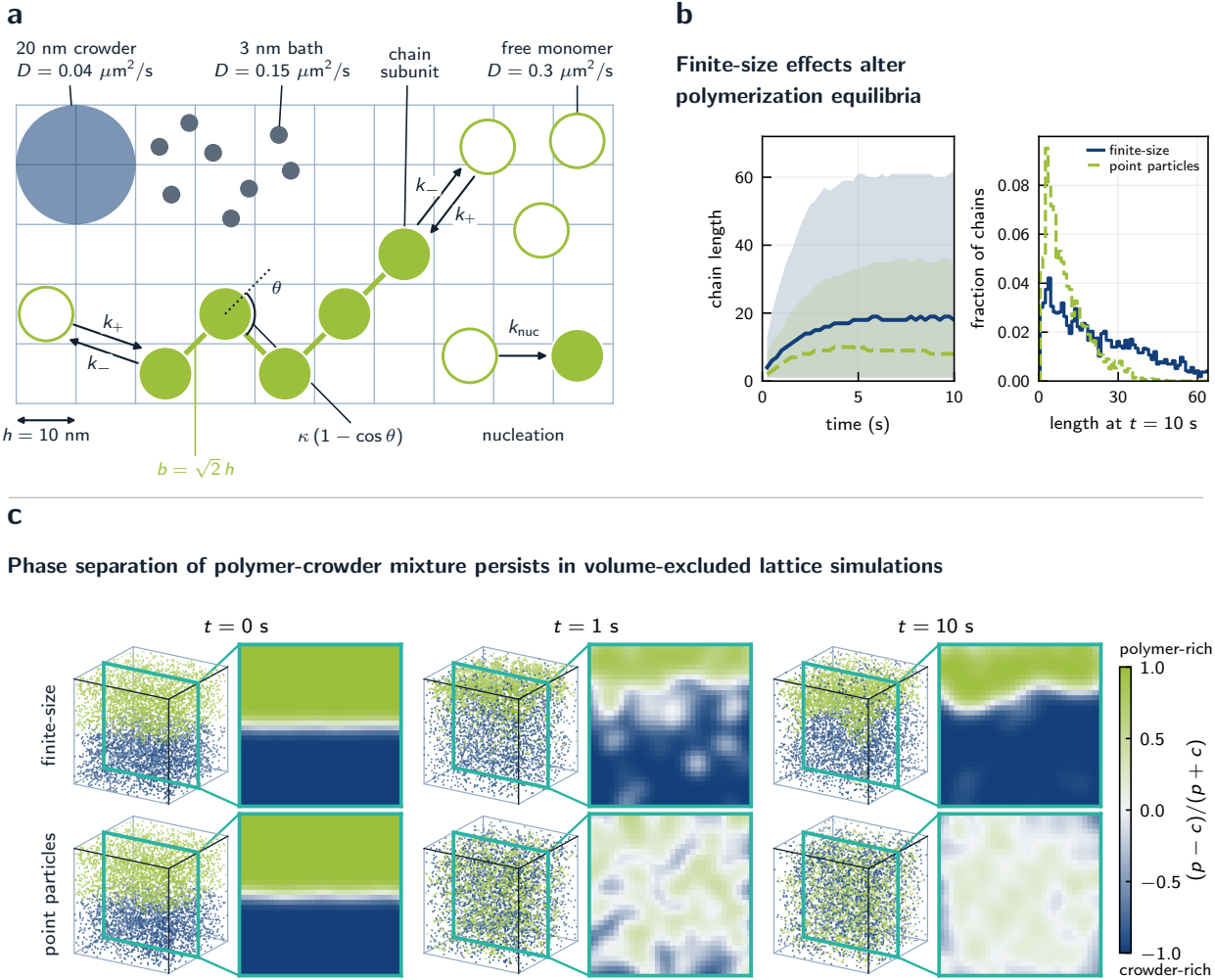

a
20 nm crowder
$D = 0.04\ \mu m^2/s$
3 nm bath
$D = 0.15\ \mu m^2/s$
chain subunit
free monomer
$D = 0.3\ \mu m^2/s$
k_-
k_+
θ
k_+
k_-
k_{nuc}
$h = 10$ nm
$\kappa\,(1 - \cos\theta)$
nucleation
$b = \sqrt{2}\,h$
b
Finite-size effects alter polymerization equilibria
chain length
time (s)
fraction of chains
length at $t = 10$ s
finite-size
point particles
c
Phase separation of polymer-crowder mixture persists in volume-excluded lattice simulations
$t = 0$ s
$t = 1$ s
$t = 10$ s
finite-size
point particles
polymer-rich
crowder-rich
$(p - c)/(p + c)$